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Experience-informed design

Information exchange to and from diverse experience research

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Abstract
Designing with experiences is integral, but when done casually, can lead to haphazard outcomes. The loss in translation of experiences from context and discipline through design research risks devaluing the significance of design research epistemologies. This research elaborates the issue of using experience as data, inputs, inspirational source in design research methods, then proposes the research needed to improve the rigour of using experience in the design process.

Keywords
Experience research, research rigour, experience-informed design

Introduction

Designers and design researchers often work in multidisciplinary teams and interact with stakeholders from different disciplines. Through these interactions, they weave one another's experiences into the design process to co-create outcomes. However, experiences are complex, mostly tacit, and difficult to articulate accurately (Dewey, 1933). This makes designing with experiences (e.g. user experience journey) and designing for end-user experience (e.g. AirBnB's UX design experience) even more challenging. Nonetheless, integrating these experiences in design is necessary, particularly when informed by diverse disciplines and attempting to create a multidisciplinary outcome (McComb & Jablokow, 2022).

Publications on experience research have exponentially multiplied four-fold in the last two decades, from 10,009 (2000 - 2010) to 39,162 (2010 - 2018)(Roto et al., 2021). As these experience research are published within their fields, experience data is collected and analysed according to the different disciplines, written from different disciplinary perspectives, and thus problematises the translation of findings across disciplines. The work of experience-driven design research draws on experience research from diverse fields for analysis of experience registers like the *Experience Sample Method* (Larson & Csikszentmihalyi, 2014), to inform the design challenge. The first challenge design researchers face is that they do not have cognate methodological knowledge of these fields (obstetrics, hospitality, gambling, AI). Adding to this is also the designers introspection (Xue & Desmet, 2019) hence the individual and co experience (Battarbee, 2003; Battarbee & Koskinen, 2005). The impetus to corroborate findings across the experience research spectrum points to collaboration by integrating, "multiple methods for triangulation, confirmation, or rejection" (Xue & Desmet, 2019, p. 57). However, there is limited, possibly no, shared language for these experience research to accurately traverse out of their disciplines and into design research. These evidence-based experience insights are untapped credible sources for design researchers.

Research approach

This paper unpacks research rigour related to sourcing and translating experience in design research by describing the current challenges of using experience in design and design research. Second, we draw on phenomenon of experiences (see e.g. Csikszentmihalyi, 1997, 2013; Dewey, 1933; Husserl, 1970; Merleau-Ponty, 1965; Turner & Bruner, 1986) to highlight experience as a critical source for design research and therefore, the impetus to develop our methodology recognising the complexity when using experience in design research. The objective is to prototype to workflow that accelerates the conversion of disciplinary-situated experience research into design research inputs, so that designers can use evidence-based experiences to enrich their design process.

Findings

We propose an input-process-output model through the lens of design research (see Figure 1) to translate experiences recorded in other literature into inputs for design. This IPO model aims to 1) articulate a process for design researchers to translate experience in research into information for design, 2) provide a descriptive method to compare how designers translate such experiences, and 3) enable the design research community to critique, test and improve the rigour of this

translation process. While we acknowledge that designers also refer to experiences outside of research literature, this IPO model aims to first leverage the abundant and rigorously documented experience data in research before expanding to incorporate sources where experience in practice is less rigorously documented (e.g., medical task analysis).

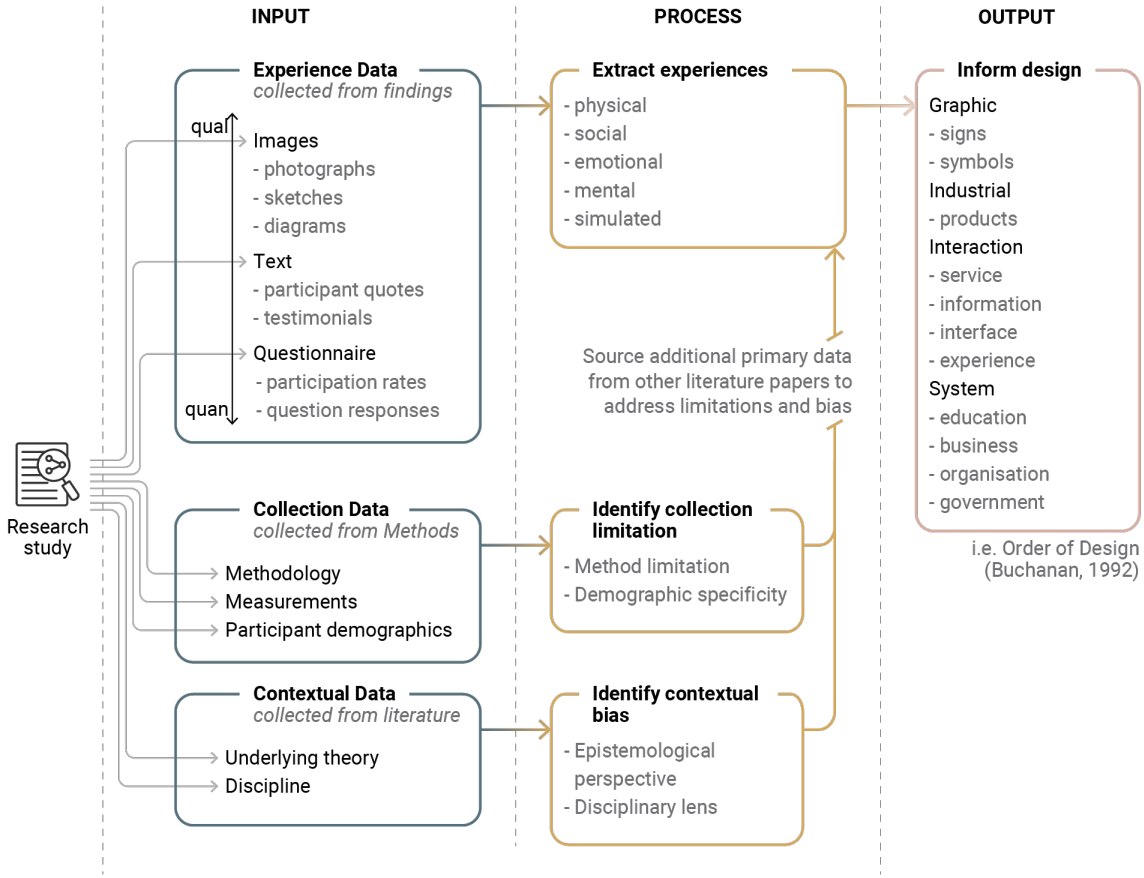


Figure 1 Experience-to-design model, A preliminary descriptive method to translate experience data into design

Inputs: Design researchers can collect three types of experience data in research articles found in other disciplines, 1) images, 2) texts and 3) questionnaire results. Images consist of photographs of participant experiences, sketches made by participants to describe their experience, and diagrams made by participants and/or researchers to visualise the abstractness of experiences. Text consists of participant quotes about experiences provided in open-ended questionnaires, or experience testimonials given by participants in interview sessions. Finally, questionnaire outcomes, such as overall responses to specific questions, may provide data on the experiences of a group of participants. Aside from these three types of experience data, design researchers may also gather information about how the experiences were collected (i.e., collection data) and how they were examined (i.e., contextual data). The purpose of collecting such metadata of the experience data is to identify the limitations of the experience data and the researcher's perspective towards examining the experience data.

Process: Next, design researchers can categorise the experience data (i.e., images, texts and questionnaire outcomes) they found in other research articles into five types of experiences, 1)

physical, 2) social, 3) emotional, 4) mental, and 5) simulated. We acknowledge that this typology is not exhaustive and require more exploration/investigation in future research. However, the significance of making these types of experiences explicit is to provide design researchers with a shared language to articulate how they translated experience in research articles into information for their design. Referencing the implicit metadata of experience data, design researchers can improve their data collection rigour by sourcing additional experience data from other articles to address collection limitations and research context biases. For example, if the design researcher only gathered experience data consisting of images and text, the researcher might consider searching for questionnaire outcomes about similar experiences. This enables the research to use both broad and in-depth experience data to inform their design.

Output: In the output stage of our IPO model, design researchers can translate the five types of experiences to inform different types of design. In this conceptual model, we draw on Buchanan's (2001) Four Orders of Design to demonstrate how experience data can be used to inform a variety of design outcomes. Again we acknowledge that there might be more suitable frameworks of describing different types of possible design outcomes. However, since the purpose of this research is to provide a preliminary descriptive model for future researchers to expand on, describing the typology of design outcomes is outside the scope of this paper.

Discussion

Our proposed Experience-to-design IPO model is a preliminary descriptive method to translate experience data into design. This model, grounded by the phenomenology of experience, serves as a research prototype (Koskinen & Frens, 2017), similar in objective to *Degrees of Disciplinarity Framework* (McComb & Jablokow, 2022) to understand their meaning or purpose (Koskinen & Frens, 2017). Experience informed design research aims to provide a portal to filter the broader experience research literature to consider (a) the role of design research as boundary object to improve experience information exchange and (b) ideally integrate experience informed design research in disciplines other than design.

Conclusion

Design research is well equipped to improve experience research knowledgeability as the field offers diverse methodologies that are scientific, creative and visual. Communicating, collating and converting qualitative experiences into quantifiable, accessible and universal forms of communication is challenging. For example drawing on methods to understand patient experiences that inform medical approaches may be applied to explicating learning experiences informing pedagogy may be made possible through design research. Experience research is a growing field that is diverse, interdisciplinary and transdisciplinary arguing the value of collective and intersectoral approaches that are experience led. Design research has a role in unpacking, distilling, clarifying and communicating diverse data presented in heterogeneous forms. The input-process-output (IPO) model is explained in this note as an experience-to-design model at the early stage of the design research process. The IPO is a preliminary descriptive method to translate experience data into design and aims to encourage diverse inputs filtered through the lens of design research. We acknowledge that the IPO model itself requires further refinement towards a

typology however early stage processes as explained by the input, process and output model help to qualify, share and disseminate experience research in the formative stages of design research.

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